MECHANICAL [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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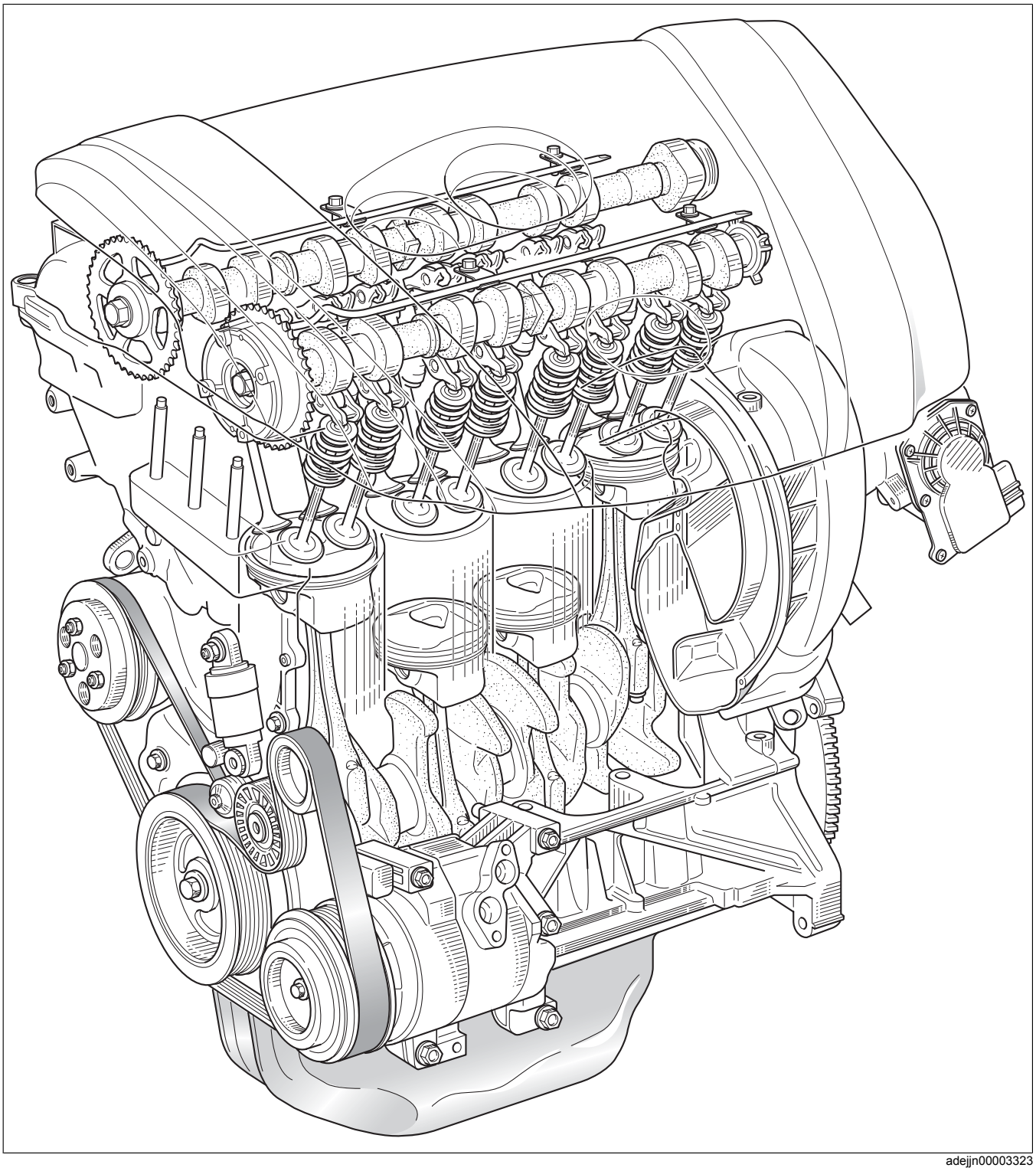
Outline

- For SKYACTIV-G 1.3 and SKYACTIV-G 1.5, the sliding resistance*1 has been reduced by the following.
  - — Adoption of a rocker arm (built in needle roller bearing)
  - — Reducing the valve spring load
  - — Narrowing down the crankshaft journal
  - — Optimizing the piston skirt shape
  - — Adoption of a low-tension piston ring
  - — Lowered tensioning of the drive belt
- A maintenance-free design for valve clearance has been achieved with the adoption of the HLA.
- With the adoption of a piston cavity, cooling loss has been reduced.
- A variable valve timing mechanism has been adopted on the intake side, reducing the pumping loss*2. (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))
- The pumping loss*2 has been reduced with the adoption of the variable valve timing mechanism on both sides of the intake and exhaust. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
- The timing chain behavior has been stabilized and sliding resistance has been reduced by optimizing the shape and rigidity of the timing chain-related parts.

*1 : Resistance (friction force) which occurs when two object slide against each other. The larger the sliding resistance, the greater the energy loss.
*2 : Energy loss which occurs from each type of resistance corresponding to intake and exhaust is called pumping loss.

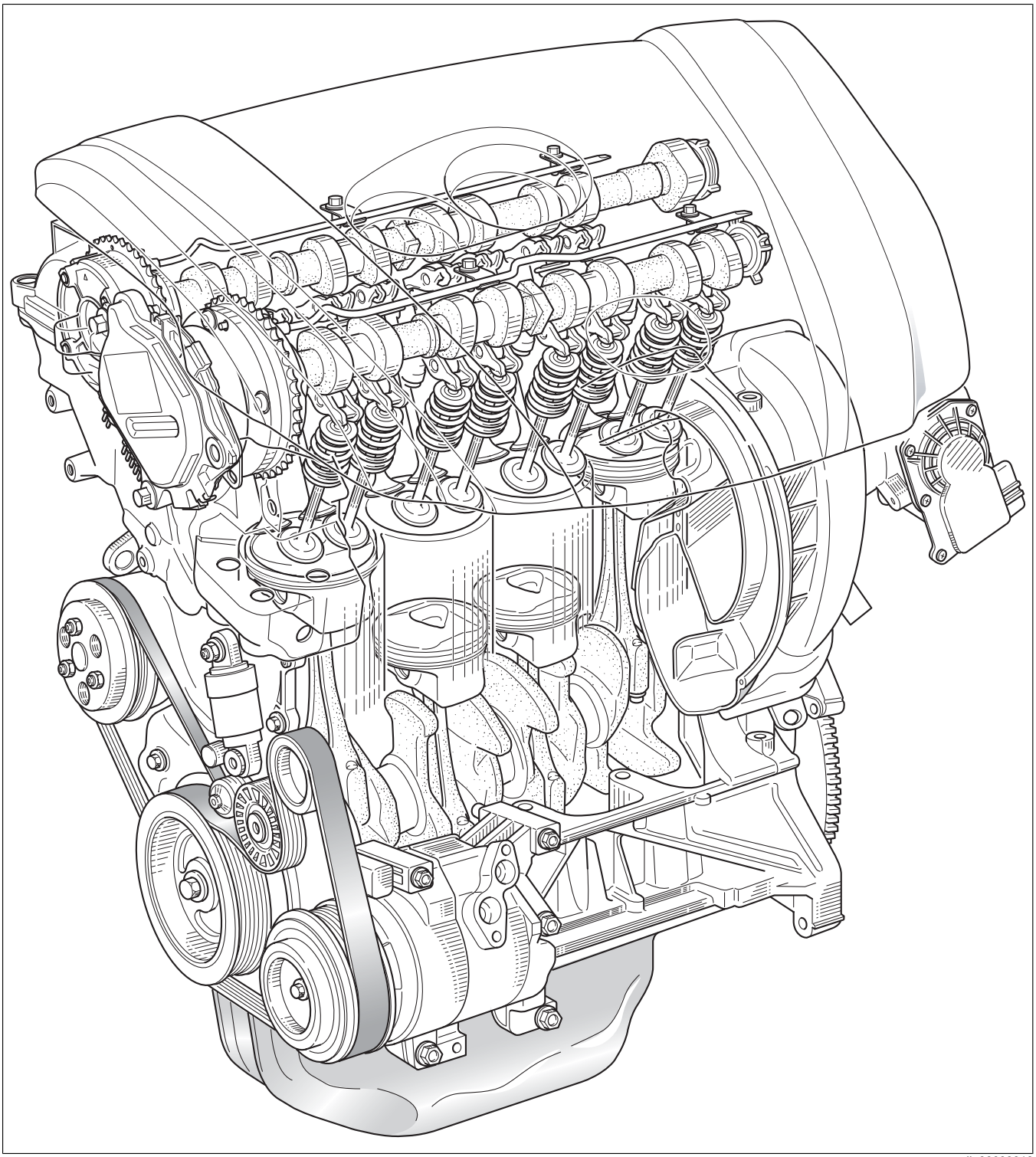
Structural View

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)



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SKYACTIV-G 1.5 (with 4-2-1 exhaust system)



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Structure

- Consists of the following parts:

Cylinder head cover	(See CYLINDER HEAD COVER [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Cylinder head	(See CYLINDER HEAD [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Cylinder head gasket	(See CYLINDER HEAD GASKET [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Cylinder block	(See CYLINDER BLOCK [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

Crankshaft		(See CRANKSHAFT, MAIN BEARING [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Piston		(See PISTON, PISTON RING, PISTON PIN [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Connecting rod		(See CONNECTING ROD, CONNECTING ROD BEARING [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Engine front cover		(See ENGINE FRONT COVER [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Crankshaft pulley		(See CRANKSHAFT PULLEY [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Drive belt		(See DRIVE BELT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Valve mechanism (See VALVE MECHANISM [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)	Valve	(See VALVE, VALVE SPRING, VALVE SEAL, VALVE GUIDE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	HLA	(See HYDRAULIC LASH ADJUSTER, ROCKER ARM [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	Camshaft	(See CAMSHAFT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Timing chain		(See TIMING CHAIN, CHAIN TENSIONER [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Variable valve timing mechanism (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)) (See VARIABLE VALVE TIMING MECHANISM [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)].)	OCV (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))	(See OIL CONTROL VALVE (OCV) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)].)
	Hydraulic variable valve timing actuator (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))	(See HYDRAULIC VARIABLE VALVE TIMING ACTUATOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)].)
Variable valve timing mechanism (SKYACTIV-G 1.5 (with 4-2-1 exhaust system)) (See VARIABLE VALVE TIMING MECHANISM [SKYACTIV-G 1.5 (WITH 4-2-1 EXHAUST SYSTEM)].)	OCV (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))	(See OIL CONTROL VALVE (OCV) [SKYACTIV-G 1.5 (WITH 4-2-1 EXHAUST SYSTEM)].)
	Hydraulic variable valve timing actuator (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))	(See HYDRAULIC VARIABLE VALVE TIMING ACTUATOR [SKYACTIV-G 1.5 (WITH 4-2-1 EXHAUST SYSTEM)].)
	Electric variable valve timing actuator (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))	(See ELECTRIC VARIABLE VALVE TIMING ACTUATOR [SKYACTIV-G 1.5 (WITH 4-2-1 EXHAUST SYSTEM)].)
Electric variable valve timing relay (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))		(See ELECTRIC VARIABLE VALVE TIMING RELAY [SKYACTIV-G 1.5 (WITH 4-2-1 EXHAUST SYSTEM)].)
Engine mount		(See ENGINE MOUNT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)